



# Data Science & Artificial Intelligence



## Artificial Intelligence

### Adversarial Search

DPP 01 Discussion Notes

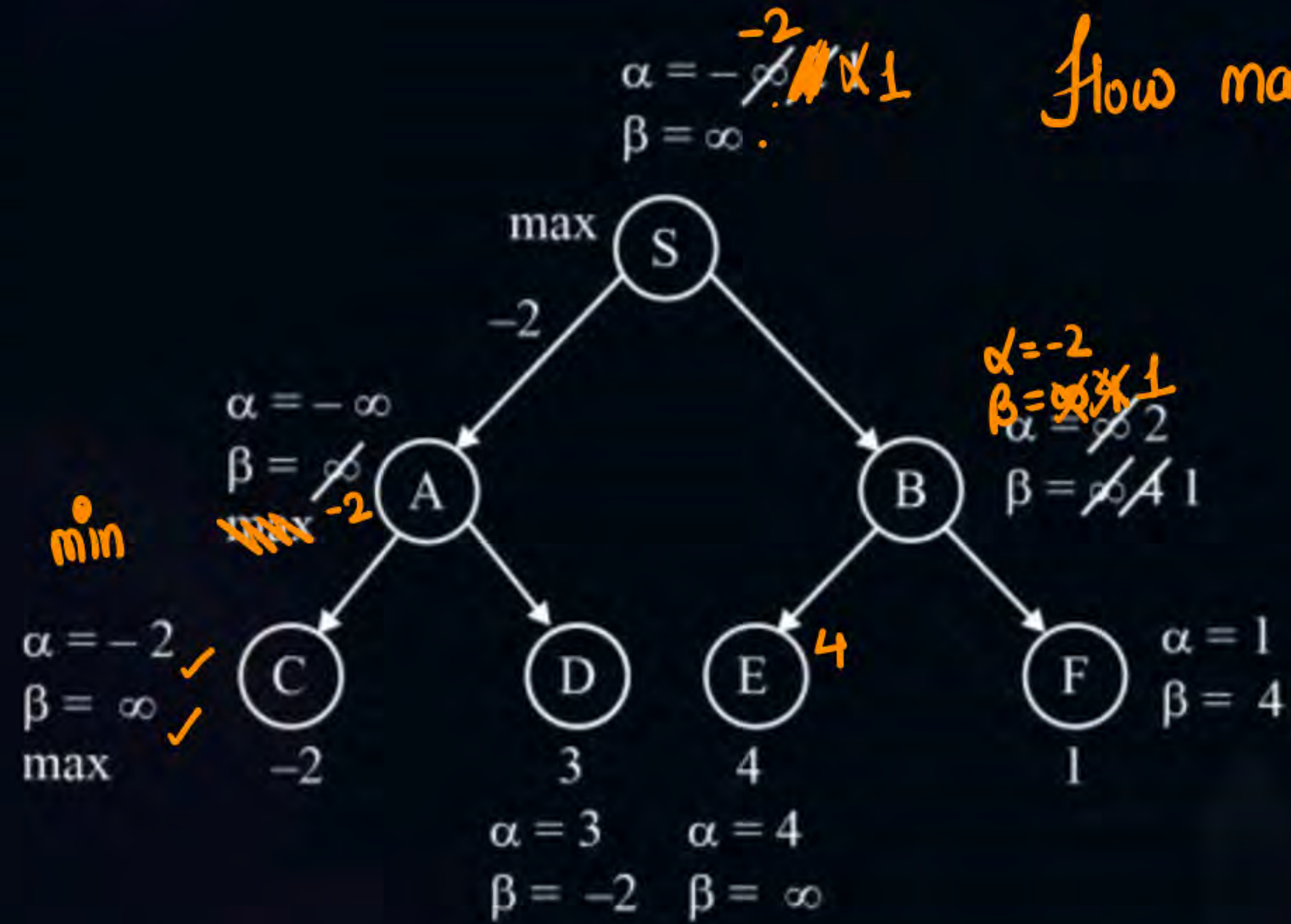


**By- Siddharth sir**

[NAT]



#Q.

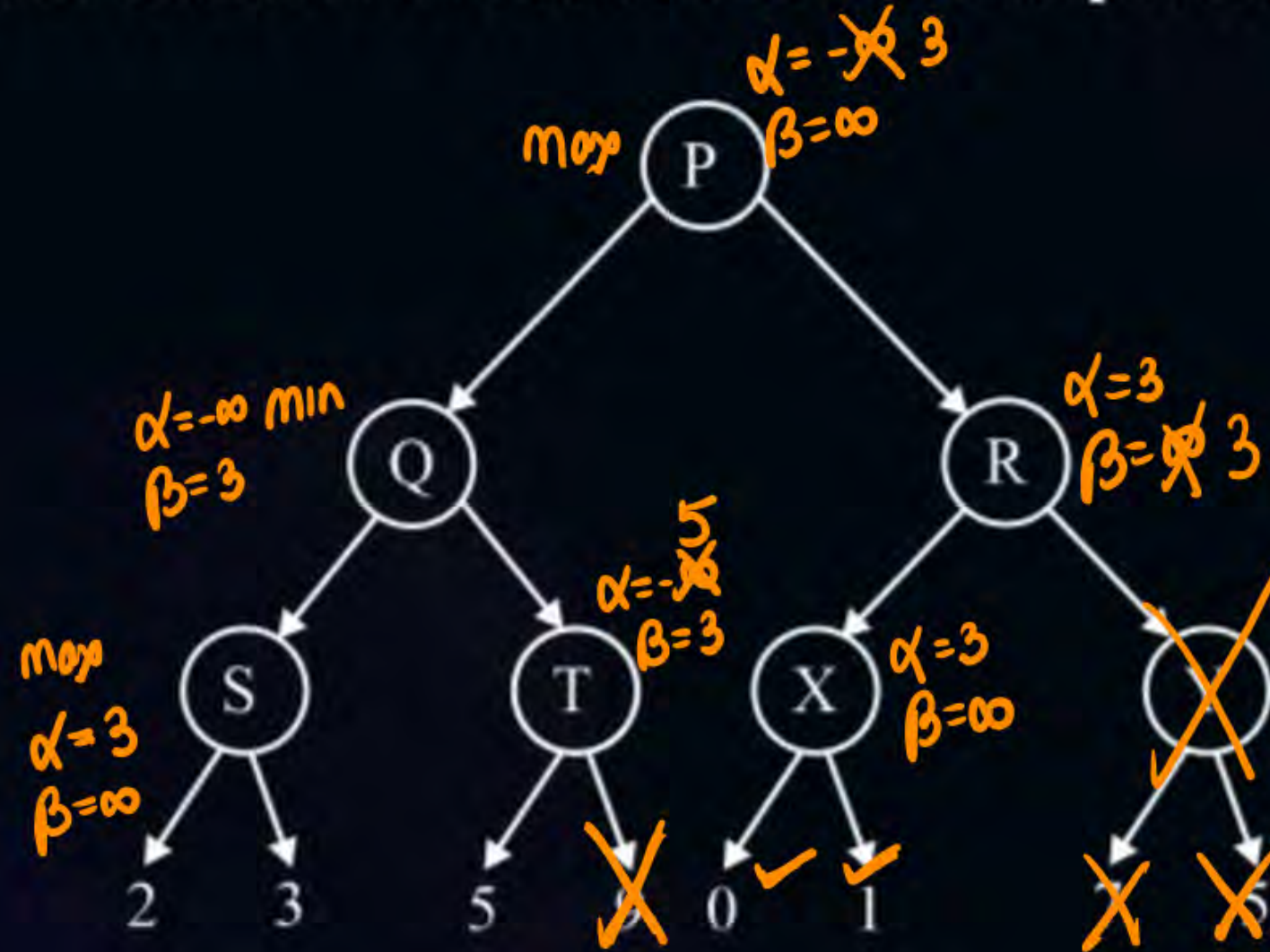


How many nodes are Pruned

$\alpha \geq \beta$  (Pruning)  
No pruning



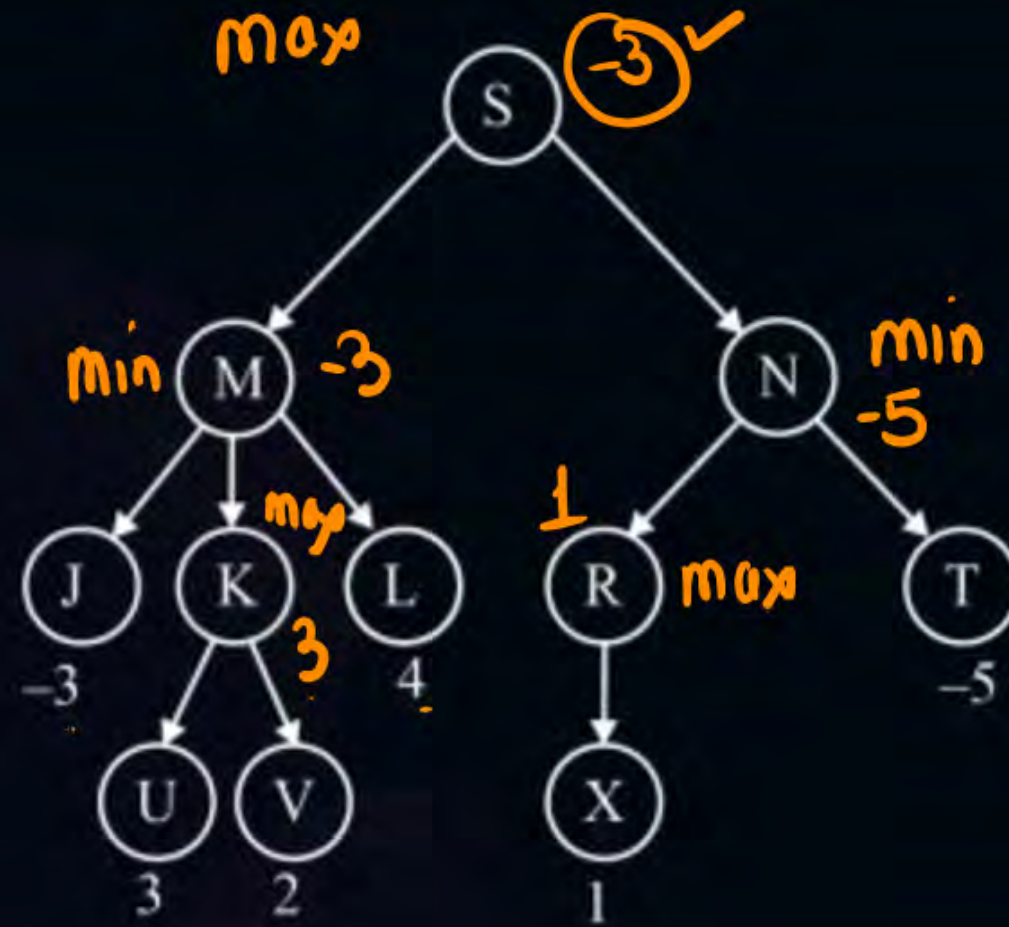
#Q. Count the number of nodes that will be pruned using  $\alpha$  -  $\beta$  pruning



Total Prune Node

Y, 3 terminal node  
④

#Q. Consider the tree green below. The root node S is the max player. What will be the best score for this root node S.

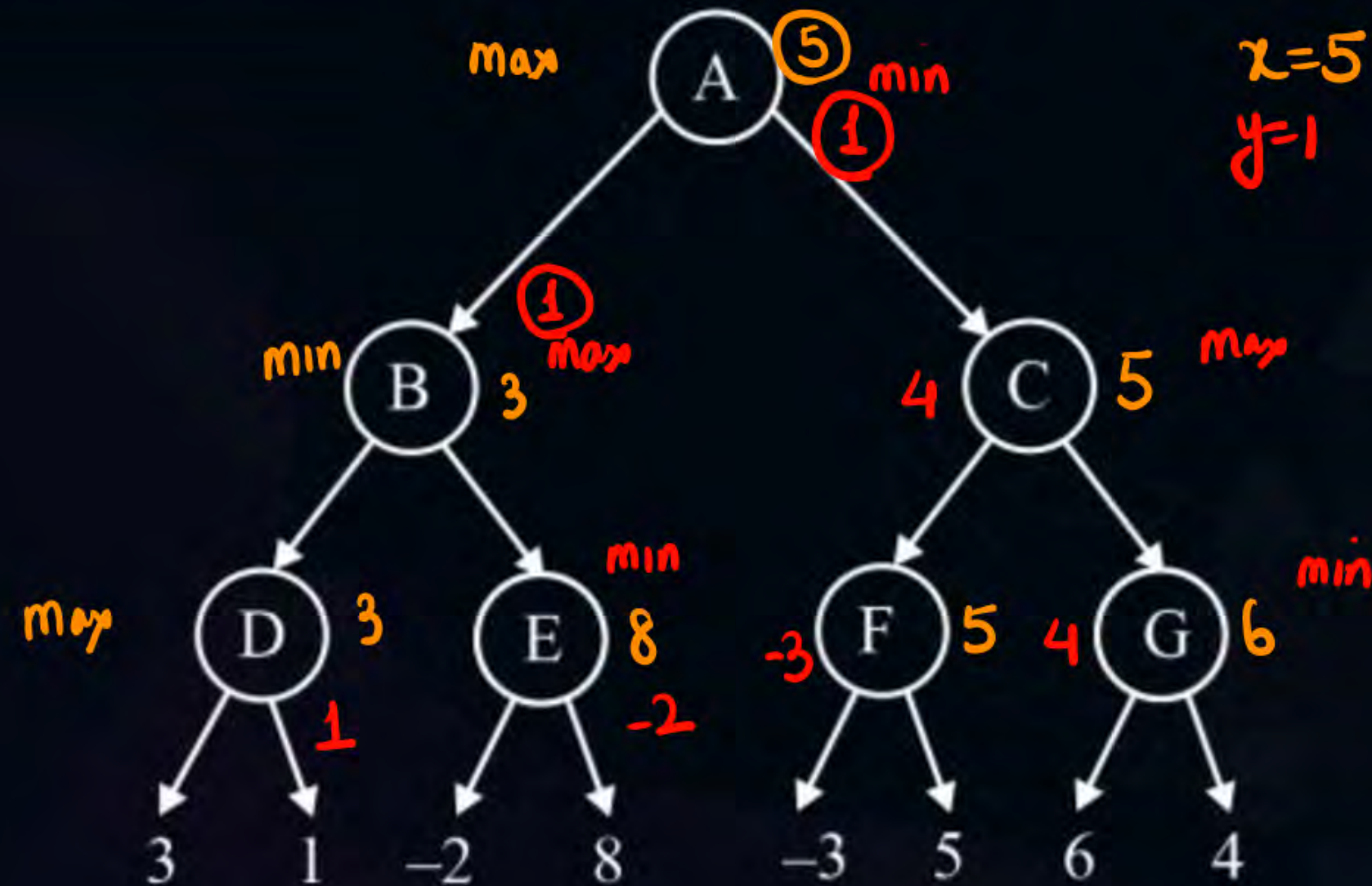




#Q. The best score possible for the root node which is a minimum player is\_\_\_\_\_.

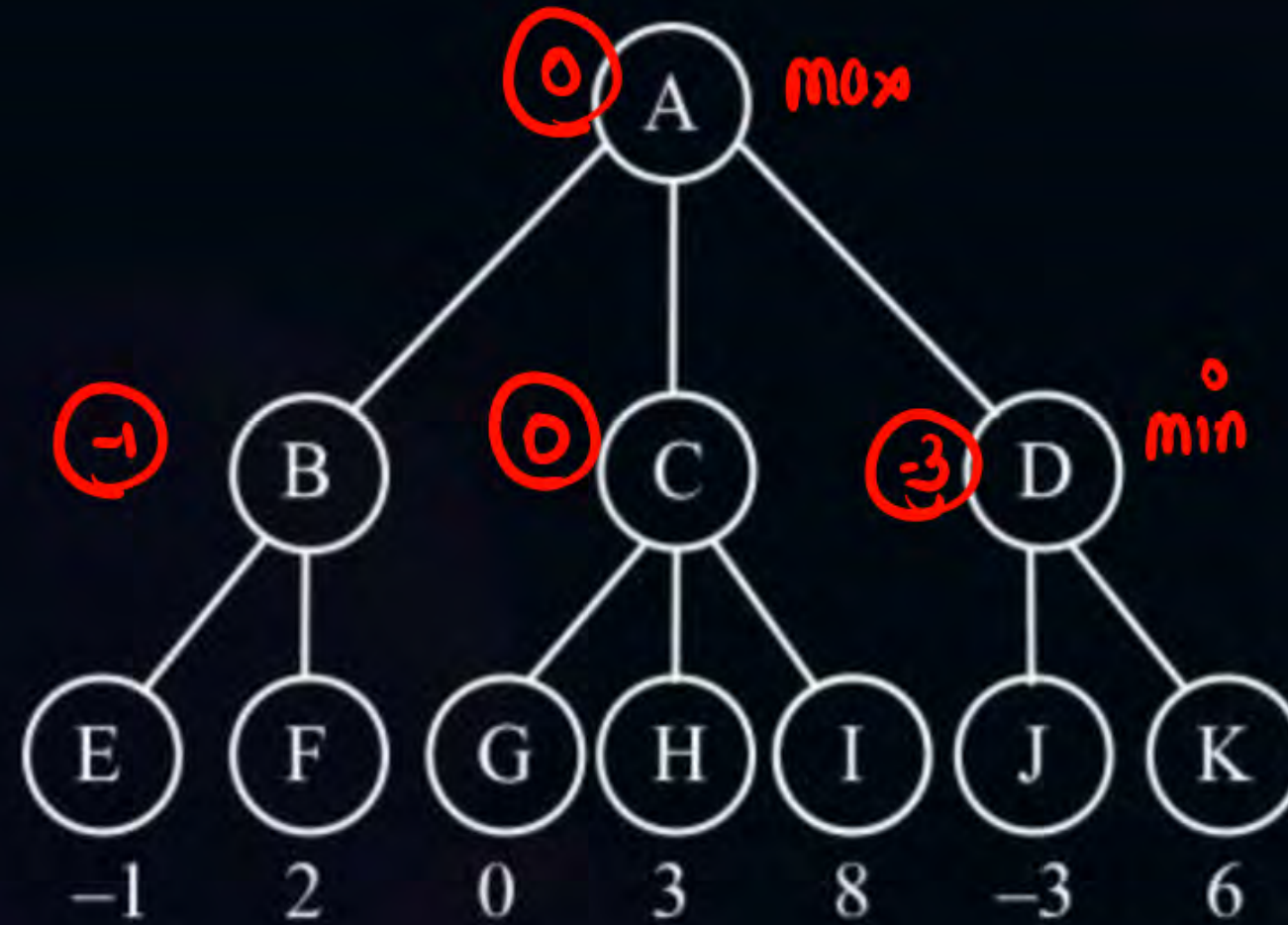


#Q. Consider the following tree. Let  $X$  denote the best score of the root node when it acts as the 'max' player. Let  $Y$  denote the best score of the root node when it acts as the 'min' player. The value of  $x + y$  is 6.

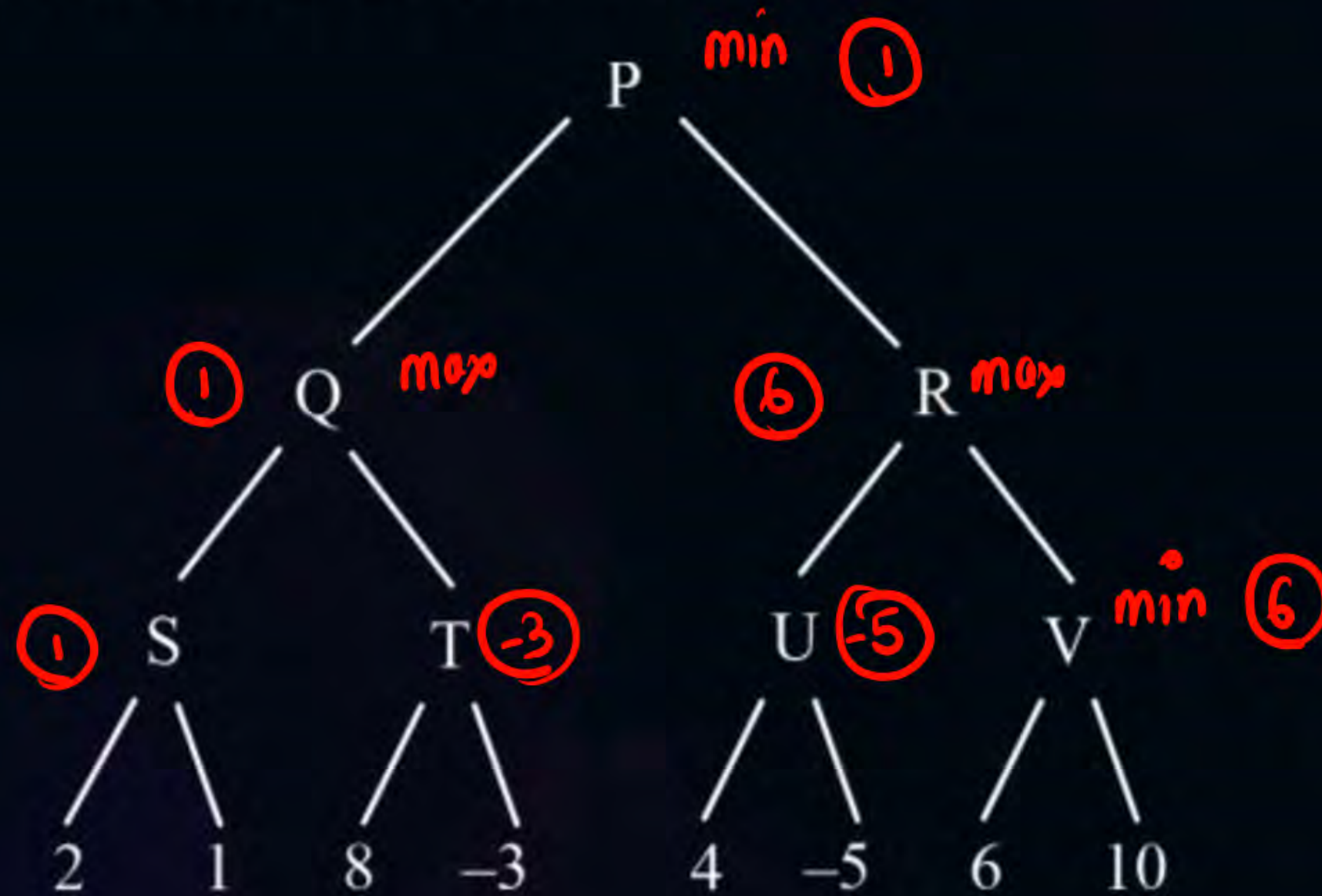




#Q. Find the best result for the root where root acts as the max player:



#Q. Find the best result for min player:

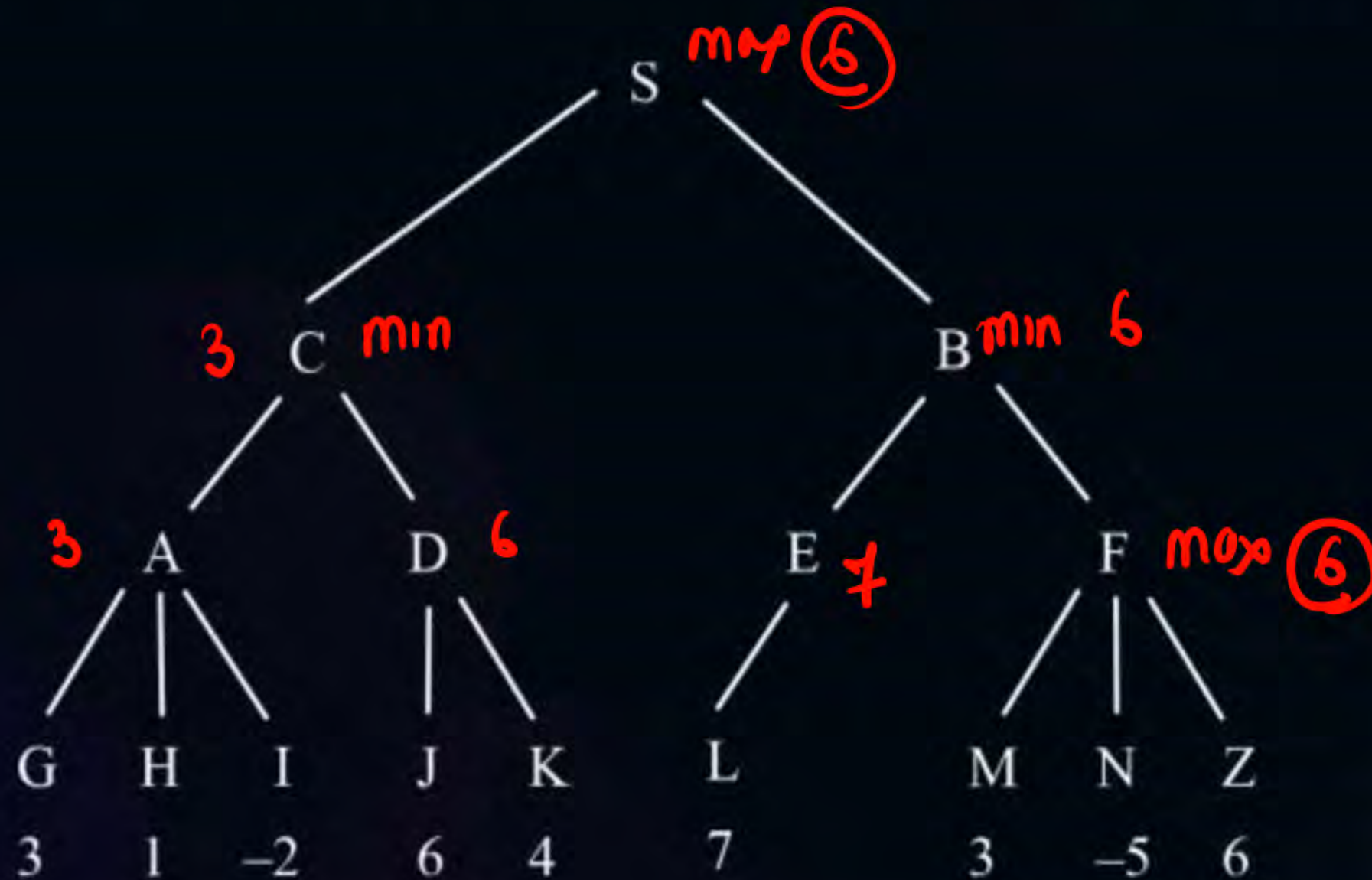




#Q. Find the best score for S (max player) in the following graph:

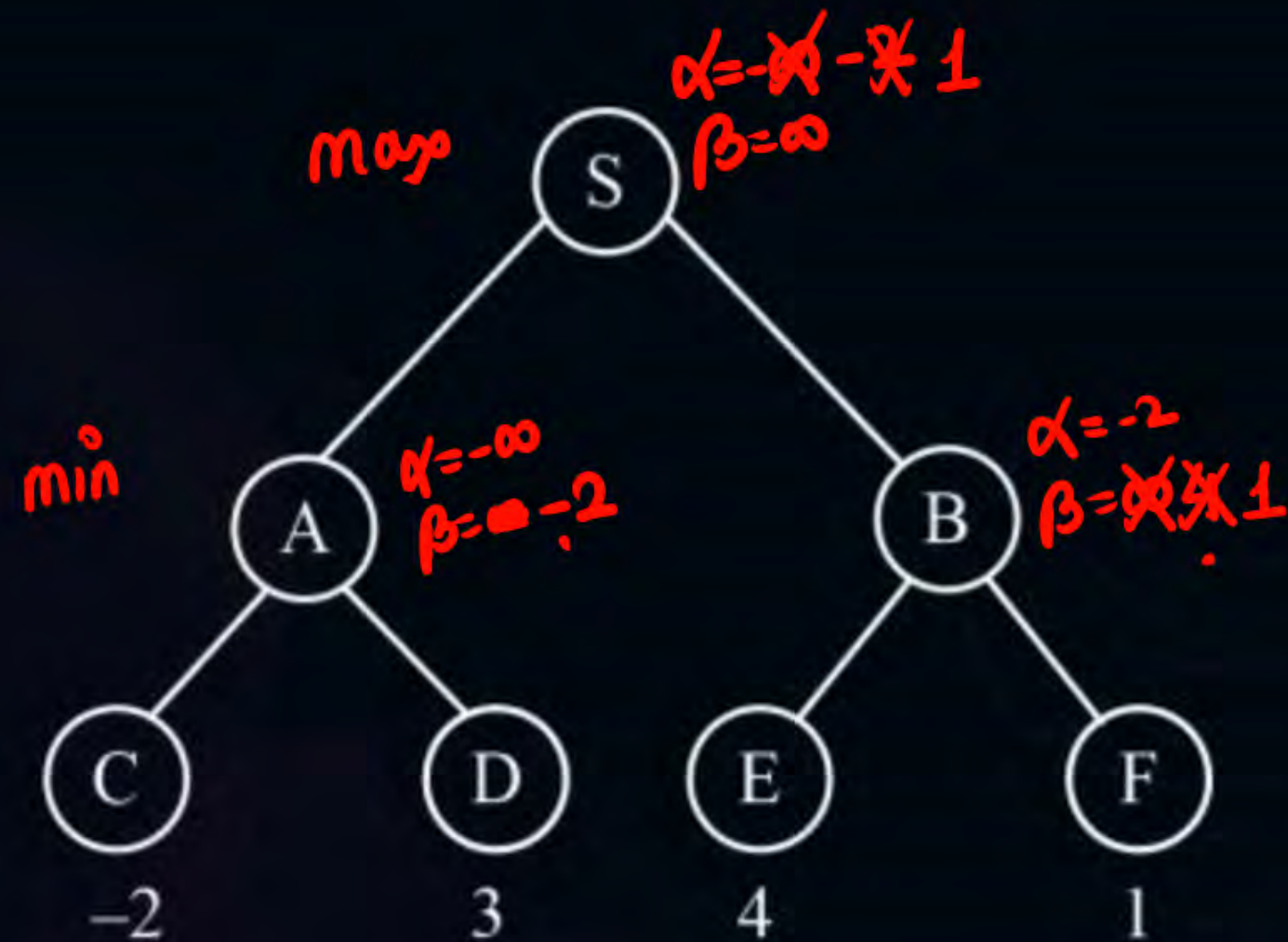


#Q. What is the best result for node C if the root node is a max player?





#Q. The number of nodes that will be pruned using alpha-beta pruning in the following graph is 0.





**THANK - YOU**